COVID-19 Wave Analysis: Growth Modelling and Clustering of Indian States Using Polynomial and Ridge Regression, PCA, K-Means and Hierarchical Clustering

Aditya Behl

Department of Computer Science, Ramanujan, University of Delhi, New Delhi, India | Roll No.: 24020570003 | E-mail: adityabehl0611@gmail.com |

Github:

- Code and data: https://github.com/Adityabehl/ml_project
- Live dashboard: https://adityabehl.github.io/ml_project/

Course: Machine Learning (B.Sc. (H) Computer Science, Semester VI) | Problem No. 13 from ML_Application_Problems.xlsx | Article type: Data Article / Data in Brief

Abstract. Indian states experienced the COVID-19 pandemic as two distinct waves that differed in timing and intensity. This study asks whether polynomial and regularised regression can describe each state's wave, and whether states can be grouped by epidemic behaviour and vaccination coverage to inform future pandemic preparedness. We curated a state-level daily panel of 22,089 records (37 state/UT codes, including the national total) from the COVID19-India static files, combining reported cases, 7-day averages, incidence per 100,000 population and CoWIN vaccination coverage. Polynomial regression (degrees 2–5) and Ridge regression (degree 4, cross-validated penalty) were fitted to 71 state–wave series. Polynomial fits gave mean R^2 = 0.826 (Wave 1) and 0.869 (Wave 2), against 0.639 and 0.388 for Ridge. Nine features for each of 35 states/UTs were standardised, reduced with PCA (first three components: 72.8% of variance) and clustered. K-means selected k = 3 (silhouette 0.300) and Ward hierarchical clustering gave a silhouette of 0.312. Two large groups had a similar Wave 2 surge (about 3.5× and 3.7×) but differed in first-dose coverage (53.3% vs 70.1%), while Mizoram formed a single-state outlier. The silhouette values indicate overlapping clusters, so the grouping is exploratory rather than definitive. The cleaned dataset, code and figures are released openly.

Keywords: COVID-19; India; epidemic waves; polynomial regression; ridge regression; PCA; k-means; hierarchical clustering; silhouette score; SDG 3; SDG 17

1 Introduction

COVID-19 reached every Indian state, but not at the same time or with the same force. The first wave (2020) and the Delta-dominated second wave (2021) differed greatly from state to state, and vaccination coverage also varied. Understanding these differences can help health departments decide where to invest in surveillance, vaccination and health-system capacity before the next outbreak. The problem is linked to SDG 3 (Good Health and Well-being) and SDG 17 (Partnerships for the Goals) [2], because it relies on open, shared data produced through a partnership of volunteers and public sources [1].

This project asks two questions. First, how well can classical regression models describe the shape of each state's daily case curve in each wave? Second, can states be grouped by their epidemic curves and vaccination coverage in a way that is interpretable? Our contributions are: (i) a curated, documented state-level panel released with code; (ii) a comparison of polynomial and Ridge regression for wave modelling using RMSE and R^2; (iii) a PCA and clustering analysis (k-means and Ward hierarchical clustering) evaluated with the silhouette score; and (iv) an illustrative SDG 3 and SDG 17 preparedness matrix.

Section 2 gives background, Section 3 describes the data, Sections 4–5 the methods and set-up, Section 6 the results, Section 7 the discussion and limitations, followed by conclusions.

2 Background

Regression finds a smooth function that describes how an outcome changes with an input; here, daily cases change with time. A straight line cannot follow a rise-and-fall wave, so polynomial regression adds powers of time to allow curvature [8]. High-degree polynomials can overfit noise, and Ridge regression reduces this by penalising large coefficients [7]. Clustering is unsupervised learning: it groups similar objects without labels. K-means [9] assigns objects to the nearest of k centres, while Ward's hierarchical method [10] repeatedly merges the pair of groups that least increases within-group variance. Principal component analysis (PCA) [11] rotates correlated features into uncorrelated components that keep as much variance as possible, which allows high-dimensional data to be plotted. The silhouette score [12] measures how well each object fits its own cluster compared with the nearest other cluster. Epidemic waves have long been described mathematically, starting with the compartmental model of Kermack and McKendrick [3]. Open dashboards such as the Johns Hopkins CSSE tracker [4] made case data widely available, and clustering of countries using COVID-19 case data has been reported [5]; general treatments of cluster analysis are given in [6]. All models were implemented with scikit-learn [13].

3 Dataset and Preprocessing

3.1 Source and description

We used two static files from the COVID19-India project (https://data.covid19india.org/): state_wise_daily.csv (daily confirmed, recovered and deceased counts by state) and cowin_vaccine_data_statewise.csv (cumulative doses administered by state, from CoWIN). The live API has been deprecated but the static files remain available; a Kaggle copy of the Indian COVID-19 data exists as an alternative [14]. The raw files are stored unchanged in data/raw/ of the released repository.

3.2 Cleaning and panel construction

The cleaning steps, all implemented in the released code, were as follows. (1) The daily case file was reshaped from wide to long format and one row per state per day was built for confirmed, recovered and deceased counts. (2) Negative daily values, which arise from retrospective reconciliation of reported counts, were set to zero. (3) A 7-day trailing rolling average of daily confirmed cases was computed to remove weekday reporting noise. (4) Cumulative confirmed and deceased counts were computed. (5) Each state was assigned a projected 2021 population, and incidence per 100,000 population was calculated for both the 7-day average and the cumulative count. (6) The vaccination file was matched to states, merged by date, and forward-filled within each state so that days without an update carry the last known value; missing values at the start were set to zero. (7) Dose 1 and Dose 2 coverage were computed as doses divided by total population, capped between 0% and 100%. The result is the curated file cleaned_state_panel.csv with 22,089 rows and 22 columns covering 37 state/UT codes, one of which is the national total.

Table 1. Main variables of the curated panel (selected from 22 columns).

Variable	Unit	Meaning
Date, State_Code, State_Name	–	Keys of the panel
Population	persons	Projected 2021 population
Confirmed_Daily	cases/day	Reported new cases (negatives set to 0)
Confirmed_7D_Avg	cases/day	7-day trailing average
Confirmed_7D_Avg_Per_100k	per 100,000	Population-adjusted 7-day average

Variable	Unit	Meaning
Cumulative_Confirmed, Cumulative_Deceased	persons	Running totals
Cumulative_Per_100k	per 100,000	Cumulative incidence
Dose1_Administered, Dose2_Administered	doses	Cumulative, forward-filled
Dose1_Coverage_Pct, Dose2_Coverage_Pct	% of population	Coverage, capped at 100

3.3 Wave definitions

Wave 1 was defined as 1 May – 31 December 2020 and Wave 2 as 1 March – 31 July 2021. A state-wave series was modelled only if it had at least 20 days of data and a peak above 5 cases per day. This left 71 state-wave series; 35 states/UTs had both waves and are used for clustering. The national total was excluded from modelling and clustering.

4 Methodology

4.1 Regression models

For each state and wave, the target is the 7-day average of daily cases and the input is time, rescaled to the interval [0, 1] within the wave so that high powers of time stay numerically stable. **Polynomial regression** fits $y = \beta0 + \beta1 \cdot t + \beta2 \cdot t^2 + \ldots + \beta d \cdot t^{\wedge} d$ by least squares. The degree d was chosen from {2, 3, 4, 5} as the one with the highest R^2. **Ridge regression** fits a degree-4 polynomial by minimising the squared error plus a penalty $\alpha \cdot \Sigma \beta^2$. The penalty strength α was chosen automatically from 20 values between 0.001 and 1000 by five-fold cross-validation (RidgeCV). Negative predictions were set to zero because case counts cannot be negative.

4.2 State-level features

Each state is described by nine features extracted from its curves and from the panel (Table 2).

Table 2. The nine features used for PCA and clustering.

Feature	Definition
Wave1_Peak_Per_100k, Wave2_Peak_Per_100k	Highest 7-day average per 100,000 in each wave

Feature	Definition
Wave1_Days_To_Peak, Wave2_Days_To_Peak	Days from the start of the wave to its peak
Delta_Surge_Ratio	(Wave 2 peak + 1) / (Wave 1 peak + 1) per 100,000
Total_Incidence_Per_100k	Maximum cumulative cases per 100,000
Case_Fatality_Rate_Pct	Cumulative deaths / (cumulative cases + 1) × 100
Dose1_Coverage_Pct, Dose2_Coverage_Pct	Coverage on the last date in the panel

4.3 PCA and clustering

Features were standardised, $z = (x - \mu)/\sigma$, so that no feature dominates because of its units. PCA was applied to the standardised matrix and the first three components were retained for plotting and interpretation. K-means was run for k = 2 to 6 (random seed 42) and the k with the highest mean silhouette score was chosen. Ward-linkage hierarchical clustering was then run with the same number of clusters and shown as a dendrogram. Cluster labels (“archetypes”) in the released code come from simple threshold rules on the mean surge and mean Dose 1 coverage; they are descriptive names, not model outputs, and this report uses neutral wording.

4.4 Evaluation metrics

RMSE = $\sqrt{[(1/n)\cdot\Sigma(y - \hat{y})^2]}$ is the typical size of the error in cases per day. $R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$ is the share of variation explained. The silhouette score for an object is $(b - a) / \max(a, b)$, where a is its mean distance to its own cluster and b its mean distance to the nearest other cluster; it ranges from −1 to 1.

4.5 Illustrative preparedness matrix

For visualisation, each state is placed on a plane with Dose 1 coverage (a proxy for SDG 17, vaccine delivery) on one axis and a resilience score (a proxy for SDG 3) on the other, where resilience = 100 − (12 × surge ratio + 15 × case fatality rate %), limited to 0–100. The weights 12 and 15 were chosen by the author for illustration and were not empirically validated. Dotted lines at 70% coverage and a score of 50 divide the plane into four zones.

5 Experimental Setup

The study is descriptive: models are fitted to the whole of each wave and no data were held out, so the reported R^2 and RMSE measure goodness of fit, not forecasting accuracy. Software: Python [x.y], scikit-learn [x.y], pandas, NumPy, SciPy, Matplotlib and seaborn on Windows; random seed 42. The full pipeline (run_pipeline.py) regenerates every table, figure and data file and completed in about 18 seconds on the author's computer. Code, data and figures are available at the repository given under Data and code availability.

6 Results

6.1 Growth modelling with polynomial and Ridge regression

Table 3. Mean regression performance over states (71 state-wave series in total).

Wave	Poly RMSE	Poly R^2	Ridge RMSE	Ridge R^2
Wave 1	309.4	0.826	515.5	0.639
Wave 2	1,280.3	0.869	3,143.9	0.388

Polynomial regression explained about 83% of the variation in Wave 1 and 87% in Wave 2, and had lower error than Ridge in both waves. RMSE values are larger in Wave 2 because Wave 2 case counts were much higher and RMSE is measured in cases per day. The Ridge fit is smoother and therefore follows multi-peak curves less closely. Fig. 2 shows fits for selected states.

As an example, for Delhi in Wave 1 (Table 4) the observed 7-day average peaked at 7,340.9 cases per day on 14 November 2020, 197 days after the start of the wave. A degree-5 polynomial captured the major rises and falls ($R^2 = 0.795$), whereas Ridge with $\alpha = 1.44$ produced a single gentle hill that missed the separate peaks ($R^2 = 0.446$).

Table 4. Delhi, Wave 1: observed peak and model fits.

Quantity	Value
Peak daily cases (7-day avg)	7,340.9 on 2020-11-14
Peak per 100,000	35.69
Days to peak	197
Polynomial	degree 5, R^2 0.795, RMSE 807.4
Ridge	α = 1.4384, R^2 0.446, RMSE 1,329.0

6.2 Principal component analysis

The first three principal components explained 38.9%, 20.7% and 13.2% of the variance, 72.8% in total; the first two together explained 59.6%. Selected loadings are shown in Table 5. PC1 is positive for both wave peaks, total incidence and vaccination coverage, so it mainly separates states with larger recorded epidemics and higher coverage (for example Goa and Kerala, on the right of the biplot) from the rest. PC2 is dominated by the surge-related features and by days to peak, and places Mizoram far above the other states. The component names shown on the dashboard are interpretive labels chosen by the author.

Table 5. Selected PCA loadings (standardised features).

Feature	PC1	PC2
Wave1_Peak_Per_100k	+0.394	−0.387
Wave2_Peak_Per_100k	+0.458	−0.080
Wave1_Days_To_Peak	+0.199	+0.361
Total_Incidence_Per_100k	+0.457	+0.072
Case_Fatality_Rate_Pct	−0.059	−0.369
Dose1_Coverage_Pct	+0.405	−0.097
Dose2_Coverage_Pct	+0.435	−0.024

6.3 Clustering

The highest silhouette score among k = 2 to 6 was obtained for k = 3 (K-means 0.2999; Ward hierarchical 0.3121). Table 6 profiles the three clusters and Appendix A lists their members.

Table 6. Cluster profiles (K-means, k = 3; values are cluster means).

Cluster	States/UTs	Delta surge ratio	Wave 2 peak per 100k	Dose 1 %	Dose 2 %
0	1 (Mizoram)	15.59	81.29	58.52	42.11
1	24	3.50	29.51	53.28	26.37
2	10	3.69	90.12	70.08	45.28

The main finding is that Clusters 1 and 2 have almost the same surge ratio (3.50 and 3.69) and differ mainly in vaccination coverage: first-dose coverage is about 17 percentage points higher and second-dose coverage about 19 points higher in Cluster 2. Cluster 2 also has a higher mean Wave 2 peak per 100,000 (90.1 against 29.5). This group is dominated by smaller states and union territories, and a possible explanation is that these places detect and report more cases per person; this explanation was not tested here. Cluster 0 contains only Mizoram, whose surge ratio of 15.59 arises largely because its Wave 1 peak was very small (4.28 per 100,000), which inflates the ratio. It is therefore a statistical outlier and not evidence of a regional pattern.

Silhouette values of about 0.30 indicate clusters that overlap. Individual states confirm this: Puducherry has a silhouette of 0.043, meaning it lies almost on the border of its cluster (Appendix B). The dendrogram (Fig. 4) shows the same structure, with Mizoram merging last.

6.4 Preparedness matrix and figures

Fig. 6 shows every state on the illustrative matrix of Section 4.5, coloured by cluster. Because the resilience score is a hand-built formula, the matrix should be read as a way of organising the results, not as a measured index. All six figures are provided in the figures/ folder at 300 DPI.

Fig. 1. National and state epidemic curves. [Insert figures/fig1_epidemic_waves_timeline.png]

Fig. 2. Polynomial versus Ridge regression fits. [Insert figures/fig2_regression_fits.png]

Fig. 3. PCA biplot of states with feature loading vectors. [Insert figures/fig3_pca_biplot.png]

Fig. 4. Hierarchical clustering dendrogram (Ward linkage). [Insert figures/fig4_hierarchical_dendrogram.png]

Fig. 5. Comparative profile of the clusters. [Insert figures/fig5_cluster_characteristics.png]

Fig. 6. Illustrative SDG 3 and SDG 17 preparedness matrix. [Insert figures/fig6_sdg_preparedness_index.png]

7 Discussion

What worked. A low-degree polynomial describes the rise and fall of a wave well enough to extract interpretable features (peak, timing, surge), and PCA shows that three components carry about 73% of the information in nine features. The cleaned panel makes it easy to repeat the analysis.

Interpretation. States separate mostly by vaccination level, not by how much worse Wave 2 was than Wave 1. This suggests that, among the features used, vaccination coverage is the main axis along which states differ after the epidemic curve is summarised. Because coverage is measured at the end of the data and not before Wave 2, the results do not show that vaccination caused a difference in outcomes.

Limitations. (i) R^2 and RMSE are in-sample, so they describe fit and not predictive ability. (ii) The polynomial degree was selected by highest R^2, which favours flexible models, while Ridge was fixed at degree 4, so the two are not compared on exactly equal terms; Ridge's cross-validation also uses folds that are not ordered in time, which may explain its lower R^2 but was not verified. (iii) Silhouette scores near 0.30 show weak cluster separation, and $k = 3$ isolates a single state. (iv) Reported cases depend on testing and reporting practice, which differ between states. (v) Populations are projected estimates, coverage is divided by the whole population including people not yet eligible, and it is a snapshot at the last date available in the static files. (vi) Case fatality uses reported deaths and cases only. (vii) The preparedness matrix uses arbitrary weights. (viii) Only 35 units are clustered, so the groups are sensitive to the features chosen. (ix) Clustering used features extracted from the curves, not the normalised curves themselves.

Ethics and SDG impact. The data are aggregate state-level counts and contain no personal information. The main risk is over-interpreting the cluster labels as judgements about state health systems, which the data do not support. Better use of open surveillance data supports SDG target 3.d (early warning and management of health risks) and SDG target 17.18 (high-quality, timely and reliable disaggregated data).

8 Conclusion and Future Work

Using open COVID19-India data, we built a documented state-level panel, showed that polynomial regression describes Wave 1 and Wave 2 curves well (mean R^2 0.83 and 0.87) while Ridge gives smoother but looser fits, and found that states group mainly by vaccination coverage, with Mizoram as an outlier. Future work will evaluate models out of sample with time-ordered cross-validation, cluster the normalised curves directly, test whether results are stable to the choice of features and number of clusters, add testing-rate data, and extend the analysis to district level.

Declarations

Data and code availability. Raw data: https://data.covid19india.org/ (accessed 06 Oct 2026; the API is deprecated but static files remain). Code, curated data, figures and results: [paste your GitHub repository link here].

Conflict of interest. The author declares no conflict of interest.

Funding. None.

Use of AI tools. The author used Claude (Anthropic), an AI assistant, to explain machine-learning concepts (regression, clustering, PCA and the evaluation metrics), to review the project code and its printed output, to prepare for the viva, and to draft the wording of this report from results that the author generated by running the released pipeline. [If any other AI tool was used to write the code or the dashboard, name it here and say what it did.] The author has checked the numbers in this report against the pipeline output and the released files, and takes full responsibility for the content.

References

[1] COVID19-India, “COVID19-India API: static CSV files,” volunteer project. [Online]. Available: https://data.covid19india.org/ (accessed 06 Oct 2026).

[2] United Nations, “Transforming our world: the 2030 Agenda for Sustainable Development,” A/RES/70/1, 2015.

[3] W. O. Kermack and A. G. McKendrick, “A contribution to the mathematical theory of epidemics,” Proceedings of the Royal Society A, vol. 115, no. 772, pp. 700–721, 1927.

[4] E. Dong, H. Du and L. Gardner, “An interactive web-based dashboard to track COVID-19 in real time,” The Lancet Infectious Diseases, vol. 20, no. 5, pp. 533–534, 2020.

[5] V. Zarikas, S. G. Poulopoulos, Z. Gareiou and E. Zervas, “Clustering analysis of countries using the COVID-19 cases dataset,” Data in Brief, vol. 31, art. 105787, 2020.

[6] L. Kaufman and P. J. Rousseeuw, Finding Groups in Data: An Introduction to Cluster Analysis. Hoboken, NJ, USA: Wiley, 1990.

[7] A. E. Hoerl and R. W. Kennard, “Ridge regression: biased estimation for nonorthogonal problems,” Technometrics, vol. 12, no. 1, pp. 55–67, 1970.

[8] T. Hastie, R. Tibshirani and J. Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd ed. New York, NY, USA: Springer, 2009.

[9] J. MacQueen, “Some methods for classification and analysis of multivariate observations,” in Proc. 5th Berkeley Symp. Math. Statist. Probab., vol. 1, 1967, pp. 281–297.

[10] J. H. Ward, “Hierarchical grouping to optimize an objective function,” Journal of the American Statistical Association, vol. 58, no. 301, pp. 236–244, 1963.

[11] I. T. Jolliffe and J. Cadima, “Principal component analysis: a review and recent developments,” Philosophical Transactions of the Royal Society A, vol. 374, art. 20150202, 2016.

[12] P. J. Rousseeuw, “Silhouettes: a graphical aid to the interpretation and validation of cluster analysis,” Journal of Computational and Applied Mathematics, vol. 20, pp. 53–65, 1987.

[13] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[14] S. Rajkumar, “COVID-19 in India,” Kaggle dataset. [Online]. Available: https://www.kaggle.com/datasets/sudalairajkumar/covid19-in-india (accessed 06 Oct 2026).

Appendix A. Cluster membership (K-means, k = 3)

Cluster	Members (state/UT codes)
0 (1)	MZ
1 (24)	AP, AR, AS, BR, CT, DN, GJ, HR, JH, JK, MH, ML, MN, MP, NL, OR, PB, RJ, TG, TN, TR, UP, UT, WB
2 (10)	AN, CH, DL, GA, HP, KA, KL, LA, PY, SK

Appendix B. Sample rows of clustering_results.csv

State	Cluster	Wave 1 peak /100k	Wave 2 peak /100k	Surge	Dose 1 %	Dose 2 %	CFR %	Silhouette
GA	2	36.05	218.75	5.93	80.99	58.44	1.89	0.305
DL	2	35.69	122.96	3.38	63.47	36.10	1.74	0.276
PY	2	32.74	117.85	3.52	46.66	25.71	1.45	0.043
KL	2	24.59	107.48	4.24	71.31	38.49	0.64	0.259
MZ	0	4.28	81.29	15.59	58.52	42.11	0.36	0.000
LA	2	29.92	71.48	2.34	70.30	51.27	0.99	0.294